NetScreen: Network-Level Screen-Time Monitoring via Encrypted Traffic Classification

TERM PROJECT SUMMARY

Dorukhan Cakir, Asmar Hasanova, Yusuf Bilal Cetinkaya, Semih Karaismailoglu

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1. Problem Statement

Network operators, enterprises, and parents need to know which applications are in use on their networks in order to enforce quality of service, usage policy, and screen-time limits. Conventional application identification relies on plaintext indicators in the traffic: the domain name in a DNS query or the Server Name Indication (SNI) field of the TLS handshake. Browsers and mobile devices now resolve names over DNS-over-HTTPS, major services such as YouTube operate over QUIC, Encrypted Client Hello conceals the server name, and a VPN encapsulates all traffic in a single tunnel. Industry systems such as Cisco NBAR2 and Encrypted Traffic Analytics respond by classifying encrypted flows from their statistical shape, namely packet sizes and timing, without decryption. NetScreen implements and evaluates a small-scale version of this approach as a screen-time monitor that operates at the network level rather than on the device, and therefore cannot be bypassed by uninstalling an application or enabling a VPN.

2. Objectives and Scope

A laptop serves as a Wi-Fi hotspot to which the team members' phones connect. Packets are captured with tshark, grouped into flows with NFStream, and summarized by features including packet-size statistics, inter-arrival times, burst length, and upload/download ratio. Three detectors operate on the same traffic: a DNS-based detector, a TLS SNI-based detector, and a machine-learning detector that uses flow shape only, with Random Forest and XGBoost on engineered features followed by a one-dimensional CNN on raw packet-size sequences. Target classes include YouTube, TikTok, Instagram, Netflix, online gaming, messaging, and video calling. A FastAPI backend maintains a per-device, per-application usage clock and serves a web dashboard. When a device exceeds its configured limit, the backend publishes an MQTT message and an ESP32 node equipped with an OLED display, LED, and buzzer issues the alert. The dataset is collected by the team on its own devices only, as fixed-length sessions per application repeated across phones, under three conditions: plain Wi-Fi, DNS-over-HTTPS enabled, and VPN enabled. The public ISCX VPN-nonVPN dataset serves as an external cross-check for the VPN condition. Protocols involved include IEEE 802.11, TCP/UDP, TLS 1.3, QUIC, DNS and DNS-over-HTTPS, VPN tunneling, and MQTT.

3. Presentation of Results

The evaluation compares the three detectors under the three network conditions. For each combination we report classification accuracy per application and per category, detection latency (time from application launch to a correct label), and false-alarm rate (incorrect alerts per hour). Results will be presented as a method-by-condition table, confusion matrices, and accuracy plots. The final presentation will include a live demonstration and a recorded video: a phone launches an application, the dashboard labels it within seconds, a VPN is enabled so that the DNS and SNI detectors lose visibility while the ML detector continues to classify, and the ESP32 node raises an alert when the usage limit is reached. Source code is maintained at github.com/anxious956/net_screen.

4. Planned Timeline

Date	Milestone
Nov 5	Capture pipeline operational; first labeled dataset; second project update
Nov 19	DNS and SNI baselines; Random Forest and XGBoost models; third project update
Dec 1 to 5	DoH and VPN experiments; 1D CNN comparison; ESP32 alert node; final presentation and demo video
Dec 11	Final report

References

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